COST OF CAPITAL
AND
CAPITAL STRUCTURE

CRITICAL QUESTIONS FOR A MANAGER?

- Where to invest? What type of projects should be considered?
- How to finance capital investment requirements? Whether manager should rely more on borrowings or equity?
- Implication of financing choices? How it impacts value of firm?
- How finance manager should raise capital for future investment requirements?

CAPITAL STRUCTURE

- A firm’s choice of how much debt it should have relative to equity is known as a capital structure decision.
- A firm’s capital structure is really just a reflection of its borrowing policy.

WHY COST OF CAPITAL IS IMPORTANT

- We know that the return earned on assets depends on the risk of those assets.
- The return to an investor is the same as the cost to the company.
- Our cost of capital provides us with an indication of how the market views the risk of our assets.
- Knowing the cost of capital can also help us determine our required return for capital budgeting projects.

COST OF CAPITAL

- Sources of fund for corporation
  - Debt- bank term loan and bond/debentures
  - Equity
  - Preference shares

COST OF EQUITY ESTIMATION

- The cost of equity is the return required by equity investors given the risk of the cash flows from the firm.
  - Business risk
  - Financial risk
- Approaches
  - The dividend growth model approach and
  - the security market line (SML) approach or CAPM

DIVIDEND GROWTH MODEL APPROACH

- the firm's dividend will grow at a constant rate, g, the price per share of the stock, P_0, can be written as:

$$P_0 = \frac{D_0 \times (1 + g)}{R_E - g} = \frac{D_1}{R_E - g}$$

where D_0 is the dividend just paid and D_1 is the next period's projected dividend.

symbol R_E (the E stands for equity) for the required return on the stock.

$$R_E = D_1/P_0 + g$$

EXAMPLE: DIVIDEND GROWTH MODEL

- Suppose that your company is expected to pay a dividend of $1.50 per share next year.
- There has been a steady growth in dividends of 5.1% per year and the market expects that to continue.
- The current price is $25. What is the cost of equity?

$$R_E = \frac{1.50}{25} + .051 = .111 = 11.1\%$$

EXAMPLE: ESTIMATING THE DIVIDEND GROWTH RATE

- One method for estimating the growth rate is to use the historical average.

Year	Dividend	Percent Change
2014	1.23	-
2015	1.30	(1.30 – 1.23) / 1.23 = 5.7%
2016	1.36	(1.36 – 1.30) / 1.30 = 4.6%
2017	1.43	(1.43 – 1.36) / 1.36 = 5.1%
2018	1.50	(1.50 – 1.43) / 1.43 = 4.9%

Average = (5.7 + 4.6 + 5.1 + 4.9) / 4 = 5.1%

EXERCISE

- Greater States Public Service, a large public utility, paid a dividend of $4 per share last year. The stock currently sells for $60 per share. You estimate that the dividend will grow steadily at a rate of 6 percent per year into the indefinite future. What is the cost of equity capital for Greater States?

$$\begin{aligned} D_1 &= D_0 \times (1 + g) \\ &= \$4 \times 1.06 \\ &= \$4.24 \end{aligned}$$

Given this, the cost of equity, R_E, is:

$$\begin{aligned} R_E &= D_1 / P_0 + g \\ &= \$4.24 / \$60 + .06 \\ &= .1307, \text{ or } 13.07\% \end{aligned}$$

The cost of equity is 13.07 percent.

g = Retention ratio (b) x Return on Equity (ROE)
b = (1- Dividend payout ratio)

COST OF EQUITY (EXTERNAL)

When a firm's future dividend payments are expected to grow at a constant rate of g per period forever, the cost of external equity, k'_e, is defined as follows:

$$k'_e = \frac{D_1}{P_{net}} + g$$

where P_{net} is the net proceeds to the firm on a per-share basis.

EXAMPLE

- consider NICOR Corporation’s current stock price is $39 per share. NICOR pays a current annual dividend of $2 per share. The consensus forecast from security analysts is that earnings and dividends will grow at an annual rate of 6.5 percent per annum for the foreseeable future.

INTERNAL VS EXTERNAL COST

The cost of internal equity capital for NICOR using a constant growth dividend valuation model is:

$$k_e = \frac{\$1.68(1+0.065)}{\$39} + 0.065$$
$$= 0.111 \text{ or } 11.1\%$$

The cost of external equity capital, assuming that new shares of stock could be sold to net the company $37 per share, is:

$$k_e' = \frac{D_1}{P_{net}} + g$$
$$= \frac{\$1.68(1+0.065)}{\$37} + 0.065$$
$$= 0.113 \text{ or } 11.3\%$$

ADVANTAGES AND DISADVANTAGES OF DIVIDEND GROWTH MODEL

- Advantage – easy to understand and use

- Disadvantages
  - Only applicable to companies currently paying dividends
  - Not applicable if dividends aren't growing at a reasonably constant rate
  - Extremely sensitive to the estimated growth rate – an increase in g of 1% increases the cost of equity by 1%
  - Does not explicitly consider risk

SML APPROACH

1. The risk-free rate, R_f.
2. The market risk premium, $E(R_M) - R_f$.
3. The systematic risk of the asset relative to average, which we called its beta coefficient, β.

Using the SML, we can write the expected return on the company's equity, $E(R_E)$, as:

$$E(R_E) = R_f + \beta_E \times [E(R_M) - R_f]$$

$$R_E = R_f + \beta_E (E(R_M) - R_f)$$

EXERCISE

- Suppose stock in Alpha Air Freight has a beta of 1.2. The market risk premium is 7 percent, and the risk-free rate is 6 percent. Alpha's last dividend was $2 per share, and the dividend is expected to grow at 8 percent indefinitely. The stock currently sells for $30. What is Alpha's cost of equity capital?

SOLUTION

$$R_E = R_f + \beta_E \times (R_M - R_f)$$
$$= .06 + 1.2 \times .07$$
$$= .144, \text{ or } 14.4\%$$

This suggests that 14.4 percent is Alpha's cost of equity.

$$R_E = D_1/P_0 + g$$
$$= \$2.16/\$30 + .08$$
$$= .152, \text{ or } 15.2\%$$

we can average them to find that Alpha's cost of equity is approximately 14.8 percent.

PROBLEM FOR STUDENTS

- Suppose your company has an equity beta of .58, and the current risk-free rate is 6.1%. If the expected market risk premium is 8.6%, what is your cost of equity capital?
- $R_E = 6.1 + .58(8.6) = 11.1\%$

ADVANTAGES AND DISADVANTAGES OF SML

- Advantages
  - Explicitly adjusts for systematic risk
  - Applicable to all companies, as long as we can estimate beta

- Disadvantages
  - Have to estimate the *expected* market risk premium, which does vary over time
  - Have to estimate beta, which also varies over time
  - We are using the past to predict the future, which is not always reliable.

EXERCISE – COST OF EQUITY

- Suppose our company has a beta of 1.5. The market risk premium is expected to be 9%, and the current risk-free rate is 6%.
- We have used analysts' estimates to determine that the market believes our dividends will grow at 6% per year and our last dividend was $2.
- Our stock is currently selling for $15.65. What is our cost of equity?
  - Using SML: $R_E = 6\% + 1.5(9\%) = 19.5\%$
  - Using DGM: $R_E = [2(1.06) / 15.65] + .06 = 19.55\%$

COST OF DEBT

- The cost of debt is the required return on our company's debt.
- Usually focus on the cost of long-term debt or bonds.
- The required return is best estimated by computing the yield-to-maturity on the existing debt.
- The cost of debt is NOT the coupon rate.
- Types of debt
  - Term loan
  - Bond/debenture

90

$c \to c/2$

$r \to r/2$

$t \to 2t$

EXAMPLE: COST OF DEBT

- Suppose we have a bond issue currently outstanding that has 25 years left to maturity.
- The coupon rate is 9%, and coupons are paid semiannually.
- The bond is currently selling for $908.72 per $1,000 bond.
- What is the cost of debt?

$$\frac{c + (F - V)/n}{(F+V)/2}$$

- Ans: YTM = 5(2) = 10% (** 5% semiannual)

$$YTM = \frac{45 + (1000 - 908.72)/50}{1908.72/2}$$

$YTM = 5\%$

↳ semiannual

EXERCISE

- Suppose the General Tool Company issued a 30-year, 7 percent bond 8 years ago. The bond is currently selling for 96 percent of its face value, or $960. What is General Tool’s cost of debt?

$$YTM = \frac{C + (F - V)/m}{(F+V)/2} = \frac{70 + (1000 - 960)/22}{980}$$

$$= 7.32\%$$

$$P = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P is the value (in rupees), n is the number of years to maturity, C is the annual coupon payment (in rupees), r is the periodic required return, M is the maturity value, and t is the time when the payment is received.

$$P = C \times \text{PVIFA}_{r,n} + M \times \text{PVIF}_{r,n}$$

Cost of debt

r= 7.373%

COST OF PREFERENCE STOCK

- The cost of preferred stock to the firm is the rate of return required by investors on preferred stock issued by the company. Because many preferred stocks are perpetuities, it is possible to use the simplified preferred stock valuation model

$$P_0 = \frac{D_p}{k_p}$$

- where is the preferred stock price P_0; , the annual preferred dividend D_p; and , the investors' required rate of return k_p. The cost of preferred stock, , is given by the following equation:

$$k_p = \frac{D_p}{P_{net}}$$

$P_{net} = 52 - 2 = 50$

$P_0 = \frac{D_0}{r}$ $r = \frac{D_0}{P_0} = \frac{4.05}{50} = 8.1\%$

EXAMPLE

- GAI has just issued 3 million shares of a preferred stock that pay an annual dividend of \$4.05 per share. The preferred stock was sold to the public at a price of \$52 per share. With issuance costs of \$2 per share, the marginal cost of preferred stock is calculated as follows:

$$k_p = \frac{\$4.05}{\$52 - \$2}$$

$$= 0.081 \text{ or } 8.1\%$$

$P_{net} = 46.4$

$D = 5$

Par value = 50

$$C_p = \frac{5 + (50 - 46.4)/15}{96.4/2}$$

EXAMPLE

- For example, Progress Energy plans an offering of \$50 par value preferred stock that will pay a \$5.00 dividend per year. The preferred stock is expected to yield Progress net proceeds of \$46.40 per share after all issue costs. The preferred stock must be retired at its par value in 15 years. The cost of this preferred stock issue can be computed by solving for k_p in the following valuation model:

$$P_{\text{net}} = \$46.40 = \$5(\text{PVIFA}_{k_p,15}) + \$50(\text{PVIF}_{k_p,15})$$

- Note: if cost of issuance of preferred stock is zero then $P_{net} = P$

COST OF PREFERENCE SHARE

- Preference capital carries a fixed rate of dividend and is redeemable in nature.

Face value	: Rs.100
Dividend rate	: 11 percent
Maturity period	: 5 years
Market price	: Rs.95

The cost of preferred stock, R_P,

$$R_P = D/P_0$$

$$\frac{11 + (100 - 95)/5}{0.4 \times 100 + 0.6 \times 95} = 12.37 \text{ percent}$$

EXERCISE

On January 2, 2018, Alabama Power Co. had two issues of ordinary preferred stock with a \$100 par value that traded on the NYSE. One issue paid \$4.64 annually per share and sold for \$99.70 per share. The other paid \$4.92 per share annually and sold for \$98.97 per share. What is Alabama Power's cost of preferred stock?

Using the first issue, we calculate that the cost of preferred stock is:

$$
\begin{aligned}
R_P &= D/P_0 \\
&= \$4.64/\$99.70 \\
&= .0465, \text{ or } 4.65\%
\end{aligned}
$$

Using the second issue, we calculate that the cost is:

$$
\begin{aligned}
R_P &= D/P_0 \\
&= \$4.92/98.97 \\
&= .0497, \text{ or } 4.97\%
\end{aligned}
$$

So, Alabama Power's cost of preferred stock appears to be about 4.81 percent.

WACC CONCEPT

- We can use the individual costs of capital that we have computed to get our “average” cost of capital for the firm.

- This “average” is the required return on the firm’s assets, based on the market’s perception of the risk of those assets.

- The weights are determined by how much of each type of financing is used.

CAPITAL STRUCTURE WEIGHTS

- Notation
  - E = market value of equity = # of outstanding shares times price per share
  - D = market value of debt = # of outstanding bonds times bond price
  - P = Market value of preferred stocks
  - V = market value of the firm = D + E + P
- Weights (if P=0 when no preference shares)
  - w_E = E/V = percent financed with equity
  - w_D = D/V = percent financed with debt

WEIGHTED AVERAGE COST OF CAPITAL (WACC)

We are concerned with aftertax cash flows, so we also need to consider the effect of taxes on the various costs of capital.

Interest expense reduces our tax liability (subject to limitation).

- This reduction in taxes reduces our cost of debt.
- After-tax cost of debt = $R_D(1-T_C)$

Dividends are not tax deductible, so there is no tax impact on the cost of equity.

WACC = $w_E R_E + w_D R_D(1-T_C)$

$$\mathbf{WACC} = (E/V) \times R_E + (P/V) \times R_P + (D/V) \times R_D \times (1 - T_C)$$

$V = E + D + P$

where R_P is the cost of preferred stock.

EXAMPLE: CAPITAL STRUCTURE WEIGHTS

- Suppose you have a market value of equity equal to $500 million and a market value of debt equal to $475 million.
  - What are the capital structure weights?
    - V = 500 million + 475 million = 975 million
    - w_E = E/V = 500 / 975 = .5128 = 51.28%
    - w_D = D/V = 475 / 975 = .4872 = 48.72%

EXTENDED EXAMPLE: WACC - I

- Equity Information
  - 50 million shares
  - $80 per share
  - Beta = 1.15
  - Market risk premium = 9%
  - Risk-free rate = 5%

- Debt Information
  - $1 billion in outstanding debt (face value)
  - Current quote = 110
  - Coupon rate = 9%, semiannual coupons
  - 15 years to maturity
- Tax rate = 21%

No. of bonds = $\frac{1\$ \text{ billion}}{\text{face value per bond}}$ = 10 million

Market value of bonds = 110 × 10 million
= $1.1 billion

$YTM = \frac{4.5 + (100-110)/30}{(100+110)/2}$

YTM = 4%

Annual r = 8%

$r_D = 8\%$

4 billion $ → equity market value

$E(R_e) = R_f + \beta(E(R_m) - R_f)$
= 15.35%

$WACC = \frac{4}{5.1} \times 15.35\% + \frac{1.1}{5.1} \times 8\% \times (1-0.21) \approx 13.40\%$

EXTENDED EXAMPLE: WACC - II

- What is the cost of equity?
  - $R_E = 5 + 1.15(9) = 15.35\%$

- What is the cost of debt?
  - N = 30; PV = -1,100; PMT = 45; FV = 1,000; CPT I/Y = 3.9268
  - $R_D = 3.927(2) = 7.854\%$

- What is the after-tax cost of debt?
  - $R_D(1-T_C) = 7.854(1-.21) = 6.205\%$

EXTENDED EXAMPLE: WACC - III

- What are the capital structure weights?
  - E = 50 million (80) = 4 billion
  - D = 1 billion (1.10) = 1.1 billion
  - V = 4 + 1.1 = 5.1 billion
  - w_E = E/V = 4 / 5.1 = .7843
  - w_D = D/V = 1.1 / 5.1 = .2157

- What is the WACC?
  - WACC = .7843(15.35%) + .2157(6.205%) = 13.38%

EXERCISE

- The FarmTrack Co. has 1.4 million shares of stock outstanding. The stock currently sells for $20 per share. The firm's debt is publicly traded and was recently quoted at 93 percent of face value. It has a total face value of $5 million, and it is currently priced to yield 11 percent. The risk-free rate is 8 percent, and the market risk premium is 7 percent. You've estimated that FarmTrack Co. has a beta of 0.74. If the corporate tax rate is 21 percent, what is the WACC of FarmTrack Co.?

$E =$ 28 million $\quad r_e = 8 + 0.74 \times 7 = 13.18\%$

$D =$ 4.65 million $\quad r_D = 11\%$

$$WACC = \frac{28}{32.65} \times 13.18 + \frac{4.65}{32.65} \times 11\% \, (0.79)$$

$= 12.54\%$

SOLUTION

- Cost of equity, using SML

  $R_E = 8\% + .74 \cdot 7\% = 13.18\%$.

- total value of the equity (E) = 1.4 million × \$20 = \$28 million.
- The debt sells for 93 percent of its face value,
- Total current market value (D)= 0.93 × \$5 million = \$4.65 million.
- total market value (V)=equity and debt together =\$28 + 4.65 = \$32.65 million.

$$\begin{aligned} \text{WACC} &= (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C) \\ &= .8576 \times .1318 + .1424 \times .11 \times (1 - .21) \\ &= .1254, \text{ or } 12.54\% \end{aligned}$$

COMPANY VALUATION WITH THE WACC

- The WACC can be useful for investment analysts when trying to measure the value of a company.
- If an analyst can predict future Cash flow from assets (CFFA) for the entire firm, WACC becomes the firm's discount rate.
- To separate financing costs from the cash flows, the tax amount should be the amount that would be paid if the firm used no debt.
- With no debt, Adjusted CFFA, or CFA*:

  CFA* = EBIT × (1 – T_C) + Depreciation – Change in NWC – Capital spending (CAPEX)
- If these cash flows continue to grow at growth rate g perpetually, the firm value today is:

  V_0 = CFA*1 / (WACC – g); CFA^*_1 is next year's projected value

FLOTATION COSTS

- The required return depends on the risk, not how the money is raised.
- However, the cost of issuing new securities should not just be ignored either.
- Basic Approach
  - Compute the weighted average flotation cost.
  - Use the target weights, because the firm will issue securities in these percentages over the long term.

WACC = 0.375 × 3 + 0.625 × 5 = 4.25% (Flotation costs)

E = x
D = 0.6x

$\frac{D}{V} = \frac{0.6}{1.6} = 0.375$

$\frac{E}{V} = 0.625$

EXAMPLE: NPV AND FLOTATION COSTS

- Your company is considering a project that will cost $1 million. The project will generate aftertax cash flows of $250,000 per year for 7 years. The WACC is 15%, and the firm's target D/E ratio is 0.6 The flotation cost for equity is 5%, and the flotation cost for debt is 3%. What is the NPV for the project after adjusting for flotation costs?

  - f_A = (.375)(3%) + (.625)(5%) = 4.25%
  - PV of future cash flows = 1,040,105
  - NPV = 1,040,105 - 1,000,000/(1-.0425) = -4,281

D/E = .6; Let E = 1; then D = .6
V = .6 + 1 = 1.6
D/V = .6 / 1.6 = .375; E/V = 1/1.6 = .625

PMT = 250,000; N = 7; I/y = 15;
PV of Project= 1,040,105
NPV=positive

$NPV = \frac{250{,}000}{0.15}\left[1 - \frac{1}{(1.15)^7}\right]$

= 10,40,104 (without flotation cost)

CAPITAL STRUCTURE

EFFECT OF FINANCIAL LEVERAGE

- Relationship Financial Leverage, EPS, and ROE

	Current	Proposed
Assets	$8,000,000	$8,000,000
Debt	$ 0	$4,000,000
Equity	$8,000,000	$4,000,000
Debt-equity ratio	0	1
Share price	$ 20	$ 20
Shares outstanding	400,000	200,000
Interest rate	10%	10%

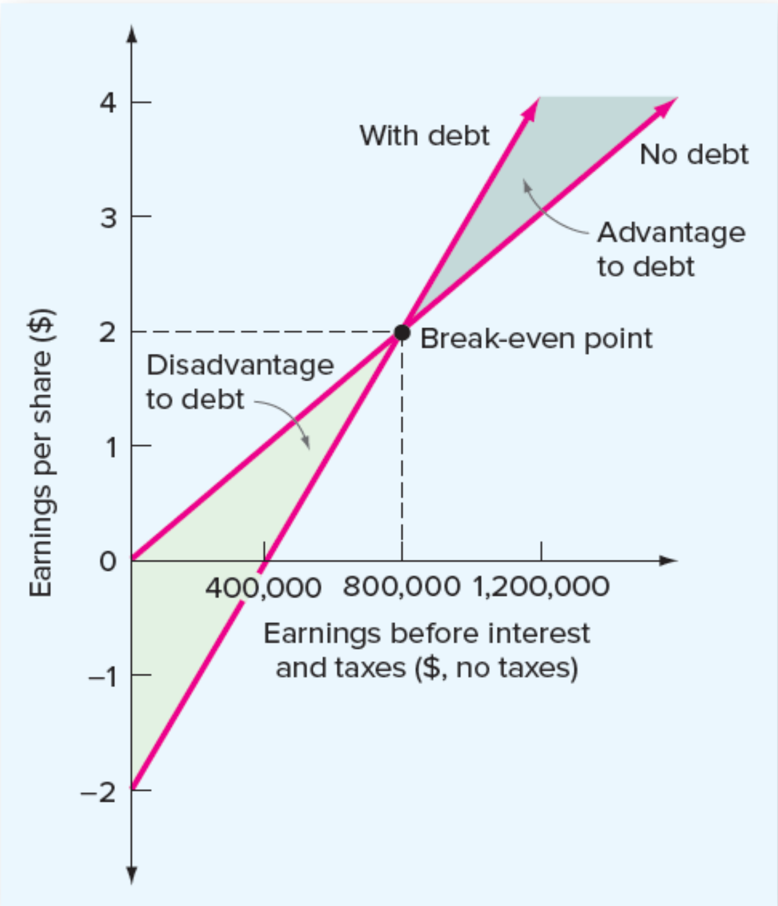
Capital Structure Scenarios

Current Capital Structure: No Debt			
	Recession	Expected	Expansion
EBIT	$500,000	$1,000,000	$1,500,000
Interest	0	0	0
Net income	$500,000	$1,000,000	$1,500,000
ROE	6.25%	12.50%	18.75%
EPS	$ 1.25	$ 2.50	$ 3.75
Proposed Capital Structure: Debt = $4 million			
EBIT	$500,000	$1,000,000	$1,500,000
Interest	400,000	400,000	400,000
Net income	$100,000	$ 600,000	$1,100,000
ROE	2.50%	15.00%	27.50%
EPS	$ .50	$ 3.00	$ 5.50

FINANCIAL LEVERAGE: EPS VS EBIT

EBIT (break even)

$$\begin{aligned} \text{EBIT}/400{,}000 &= (\text{EBIT} - \$400{,}000)\,/200{,}000 \\ \text{EBIT} &= 2 \times (\text{EBIT} - \$400{,}000) \\ &= \$800{,}000 \end{aligned}$$



EXERCISE FOR STUDENTS

- The MPD Corporation has decided in favor of a capital restructuring. Currently, MPD uses no debt financing. Following the restructuring, debt will be $1 million. The interest rate on the debt will be 9 percent. MPD currently has 200,000 shares outstanding, and the price per share is $20. If the restructuring is expected to increase EPS, what is the minimum level for EBIT that MPD's management must be expecting? Ignore taxes in answering.

$E = 4$ million

$D = 1$ million

$E = 0.8$

$D = 0.2$ $\quad r_e = 9\%$

Interest $= 90{,}000$

$$EPS_{earlier} = \frac{EBIT}{200{,}000}$$

$$EPS_{new} = \frac{EBIT - 90{,}000}{150{,}000}$$

SOLUTION

Under the old capital structure, EPS is EBIT/200,000. Under the new capital structure, the interest expense will be \$1 million × .09 = \$90,000.

$$\begin{aligned} \text{EBIT}/200{,}000 &= (\text{EBIT} - \$90{,}000)/150{,}000 \\ \text{EBIT} &= 4/3 \times (\text{EBIT} - \$90{,}000) \\ &= \$360{,}000 \end{aligned}$$

Verify that, in either case, EPS is \$1.80 when EBIT is \$360,000. Management at MPD is apparently of the opinion that EPS will exceed \$1.80.

IMPLICATION: EBIT (BREAKEVEN)

1. The effect of financial leverage depends on the company's EBIT. When EBIT is relatively high, leverage is beneficial.
2. Under the expected scenario, leverage increases the returns to shareholders, as measured by both ROE and EPS.
3. Shareholders are exposed to more risk under the proposed capital structure because the EPS and ROE are much more sensitive to changes in EBIT in this case.
4. Because of the impact that financial leverage has on both the expected return to stockholders and the riskiness of the stock, capital structure is an important consideration.

CAPITAL STRUCTURE THEORIES

- Net Income Approach
- Net Operating Income Approach
- Traditional Approach
- Modigliani and Miller (M&M)Theory of Capital Structure
  - Proposition I – Firm value
  - Proposition II – Cost of capital

NET INCOME APPROACH
(DAVID DURAND'S 1959)

- Assumptions
  - The increase in debt will not affect the confidence levels of the investors.
  - There are only two sources of finance; debt and equity. There are no sources of finance like Preference Share Capital and Retained Earnings.
  - All companies have a uniform dividend payout ratio; it is 1.
  - There is no flotation cost, no transaction cost, and corporate dividend tax.
  - The capital market is perfect; it means information about all companies is available to all investors, and there are no chances of overpricing or underpricing of security. Further, it means that all investors are rational. So, all investors want to maximize their return by minimizing risk.
  - All sources of finance are for infinity. There are no redeemable sources of finance.

NET INCOME APPROACH
(DURAND'S THEORY)

$$r_A = r_D\left[\frac{D}{D+E}\right] + r_E\left[\frac{E}{D+E}\right]$$

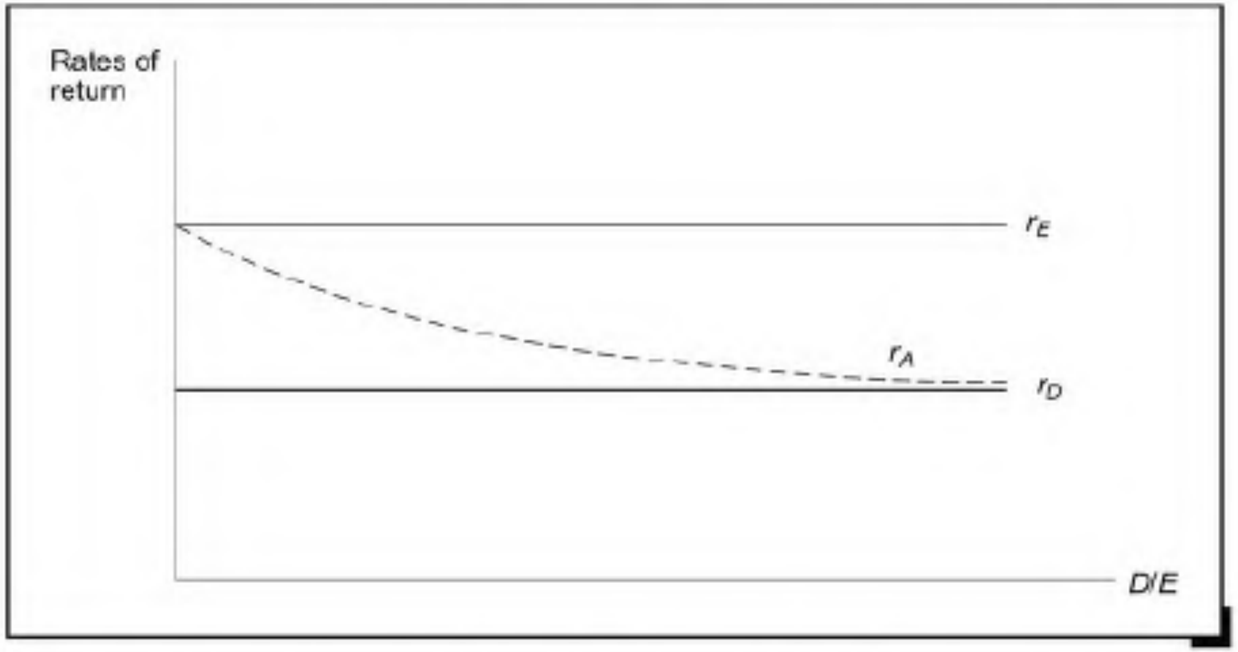
		Firm A	Firm B
O	Operating income[1]	Rs. 10,000	Rs. 10,000
I	Interest on debt	Rs. 0	Rs. 3,000
P	Equity earnings	Rs. 10,000	Rs. 7,000
r_E	Cost of equity capital	10%	10%
r_D	Cost of debt capital	6%	6%
E	Market value of equity	Rs.100,000	Rs. 70,000
D	Market value of debt	Rs. 0	Rs. 50,000
V	Total value of the firm	Rs.100,000	Rs.120,000

The average cost of capital for firm A is:

$$6\% \times \frac{0}{100,000} + 10\% \times \frac{100,000}{100,000} = 10\%$$

The average cost of capital for firm B is:

$$6\% \times \frac{50,000}{120,000} + 10\% \times \frac{70,000}{100,000} = 8\frac{1}{3}\%$$



NET OPERATING INCOME APPROACH

According to the net operating income approach, the overall capitalisation rate and the cost of debt remain constant for all degrees of leverage. In the equation

$$r_A = r_D\left[\frac{D}{D+E}\right] + r_E\left[\frac{E}{D+E}\right]$$

r_A and r_D are constant for all degrees of leverage. Given this, the cost of equity can be expressed as:

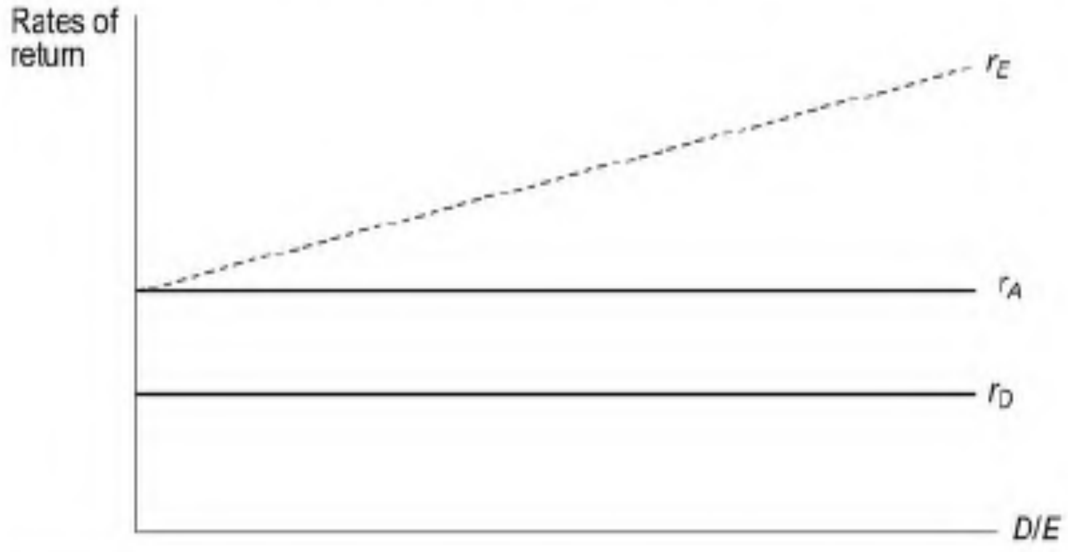
$$r_E = r_A + (r_A - r_D)(D/E)$$

$$r_A = r_D\left[\frac{D}{D+E}\right] + r_E\left[\frac{E}{D+E}\right]$$

$$r_E\left[\frac{E}{D+E}\right] = r_A - r_D\left[r_D\frac{D}{D+E}\right]$$

$$r_E\left[\frac{E}{D+E}\right]\left[\frac{D+E}{E}\right] = r_A\left[\frac{D+E}{E}\right] - r_D\left[\frac{D}{D+E}\right]\left[\frac{D+E}{E}\right]$$

$$r_E = r_A + (D/E)(r_A - r_D)$$



EXAMPLE

		Firm A	Firm B
O	Net operating income	10,000	10,000
r_A	Overall capitalisation rate	0.15	0.15
V	Total market value	66,667	66,667
I	Interest on debt	1,000	3,000
r_D	Debt capitalisation rate	0.10	0.10
D	Market value of debt	10,000	30,000
E	Market value of equity	56,667	36,667
D/E	Debt-equity ratio	0.176	0.818

The equity capitalisation rates of firms A and B are as follows:

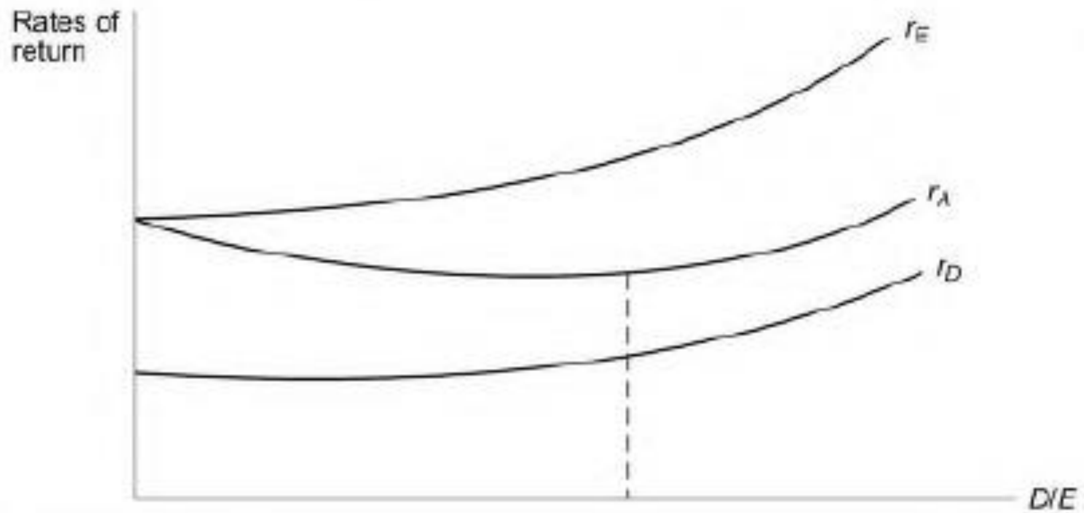
$$\text{Firm } A = \frac{\text{Equity earnings}}{\text{Market value of equity}} = \frac{9{,}000}{56{,}667} = 0.159 = 15.9\%$$

$$\text{Firm } B = \frac{\text{Equity earnings}}{\text{Market value of equity}} = \frac{7{,}000}{36{,}667} = 0.191 = 19.1\%$$

TRADITIONAL APPROACH
(EZRA SOLOMAN 1963)

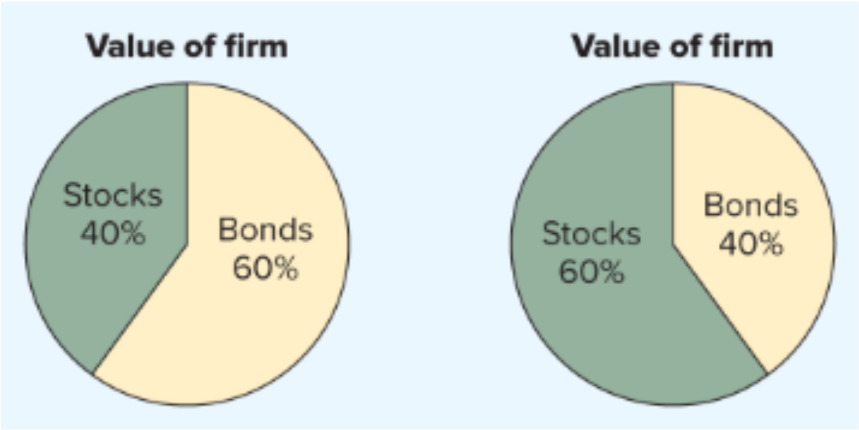
The main propositions of the traditional approach are:

1. The cost of debt capital, r_D, remains more or less constant up to a certain level of leverage but rises thereafter at an increasing rate.
2. The cost of equity capital, r_E, remains more or less constant or rises only gradually up to a certain level of leverage and rises sharply thereafter.
3. The average cost of capital, r_A, as a consequence of the above behaviour of r_E and r_D, (i) decreases up to a certain point; (ii) remains more or less unchanged for moderate increases in leverage thereafter; and (iii) rises beyond a certain point.



MM PROPOSITIONS

- **M&M PROPOSITION I: THE PIE MODEL**
- M&M Proposition I: The proposition that the value of the firm is independent of the firm's capital structure.
- Their assets and operations are exactly the same.



WHY MM THEORY IS RELEVANT?

- ideal world, the total market value of all the securities issued by a firm would be governed by the earning power and risk of its underlying real assets and would be independent of how the mix of securities issued to finance it was divided between debt instruments and equity capital.

$$\text{WACC} = (E/V) \times R_E + (D/V) \times R_D$$

where $V = E + D$. We also saw that one way of interpreting the WACC is as the required return on the firm's overall assets.

$$R_A = (E/V) \times R_E + (D/V) \times R_D$$

MM: BUSINESS AND FINANCIAL RISK

Business risk: The equity risk that comes from the nature of the firm's operating activities.
Financial risk: The equity risk that comes from the financial policy (the capital structure) of the firm.

$$R_E = R_A + (R_A - R_D) \times (D/E) \text{ by M\&M Proposition II}$$

The first component, R_A, is the required return on the firm's assets overall, and it depends on the nature of the firm's operating activities. The risk inherent in a firm's operations is called the business risk of the firm's equity.

The second component in the cost of equity, $(R_A - R_D) \times (D/E)$, is determined by the firm's financial structure. For an all-equity firm, this component is zero. As the firm begins to rely on debt financing, the required return on equity rises. This occurs because the debt financing increases the risks borne by the stockholders. This extra risk that arises from the use of debt financing is called the financial risk of the firm's equity.

HOMEMADE LEVERAGE & MM THEORY

- The shareholders can adjust the amount of financial leverage by borrowing and lending on their own. This use of personal borrowing to alter the degree of financial leverage is called homemade leverage.
- “The use of personal borrowing to change the overall amount of financial leverage to which the individual is exposed.”

EXAMPLE: HOMEMADE LEVERAGE

	Firm U	*Firm L*
Operating income (EBIT)	150,000	150,000
Interest	0	60,000
Equity earnings	150,000	90,000
Cost of equity	0.15	0.16
Market value of equity	1,000,000	562,500
Coost of debt	–	0.12
Market value of debt	0	500,000
Market value of the firm	1,000,000	1,062,500
Average cost of capital	0.15	0.1412

Such a situation, argue MM, cannot persist because equity investors would do well to sell their equity in firm L (the firm which is valued more) and invest in firm U (the firm which is valued less) with personal leverage.

For example, if an investor owns 10 percent equity in firm L, he would do well to:

1. Sell his equity in firm *L* for Rs.56,250.
2. Borrow Rs.50,000, an amount equal to 10 percent of *L*'s debt, at an interest rate of 12 percent.
3. Buy 10 percent of firm *U*'s equity for Rs.100,000.

Note that while he collects Rs. 106,250 (Rs. 56,250 as sale proceeds of his equity plus Rs. 50,000 as borrowings), his investment is only Rs. 100,000, leaving him with a surplus amount of Rs. 6,250. Yet his income remains the same:

	Old income from investment in L	*New income from investment in U*
■ 10% of firm's equity income	9,000	15,000
■ 12% interest on Rs. 50,000 loan		(6,000)
	9,000	9,000

If he invests the surplus amount of Rs. 6,250 also, his new income will be greater than his old income. Note that investors risk exposure remains unchanged because he has merely replaced Rs. 50,000 of personal borrowing for his share of firm L's corporate borrowing.

EFFECT OF:
HOME MADE LEVERAGE

- When investors sell their equity in firm L and buy the equity in firm U with personal leverage, the market value of firm L tends to decline and the market value of firm U tends to rise. This process continues until the market values of both the firms become equal because only then the possibility of earning a higher income, for a given level of investment and leverage, by arbitraging is eliminated. As a result, the cost of capital for both the firms becomes the same.

THE COST OF EQUITY AND FINANCIAL LEVERAGE

M&M PROPOSITION II

- The proposition that a firm's cost of equity capital is a positive linear function of the firm's capital structure.
- Although changing the capital structure of the firm does not change the firm's total value, it does cause important changes in the firm's debt and equity.

$$\text{WACC} = (E/V) \times R_E + (D/V) \times R_D$$

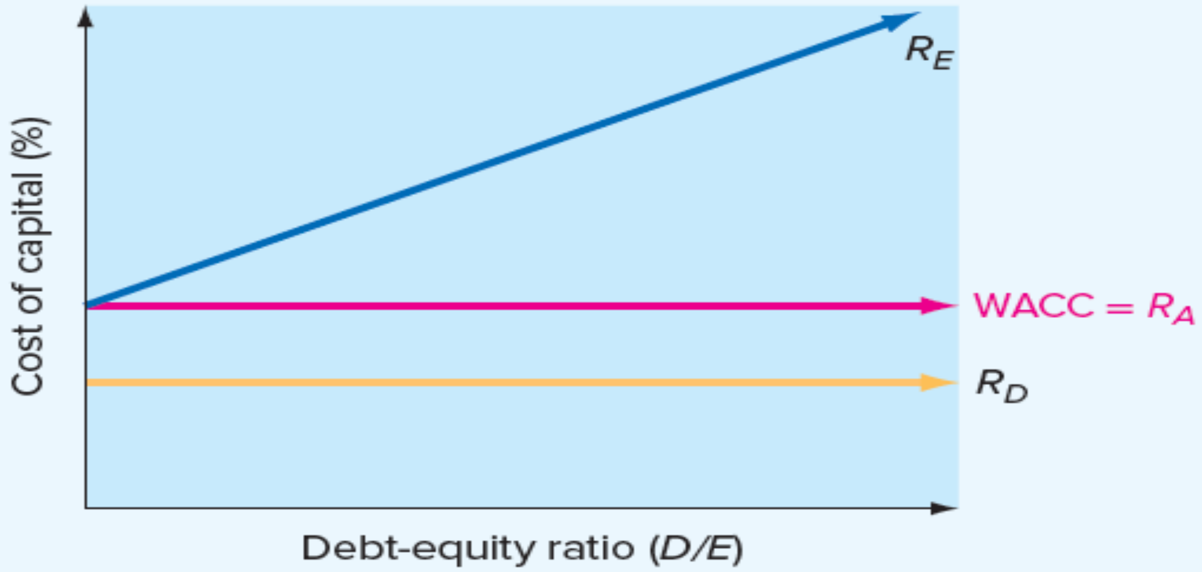
$$R_A = (E/V) \times R_E + (D/V) \times R_D$$

If we rearrange this to solve for the cost of equity capital,

$$R_E = R_A + (R_A - R_D) \times (D/E)$$

EXPLANATION

- M&M Proposition II, which tells us that the cost of equity depends on three things: The required rate of return on the firm’s assets, RA; the firm’s cost of debt, RD; and the firm’s debt-equity ratio, D/E.



$R_E = R_A + (R_A - R_D) \times (D/E)$ by M&M Proposition II

$R_A = \text{WACC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D$

where $V = D + E$

M&M PROPOSITIONS I AND II WITH CORPORATE TAXES

- Debt two distinguishing features
  - interest paid on debt is tax deductible (good for the firm)
  - failure to meet debt obligations can result in bankruptcy (not good for the firm)
- Assumption relaxed: all interest paid is tax deductible

EXERCISE

- The Ricardo Corporation has a weighted average cost of capital (ignoring taxes) of 12 percent. It can borrow at 8 percent. Assuming that Ricardo has a target capital structure of 80 percent equity and 20 percent debt, what is its cost of equity? What is the cost of equity if the target capital structure is 50 percent equity? Calculate the WACC using your answers to verify that it is the same.

SOLUTION

According to M&M Proposition II, the cost of equity, R_E, is:

$$R_E = R_A + (R_A - R_D) \times (D/E)$$

In the first case, the debt-equity ratio is .2/.8 = .25, so the cost of the equity is:

$$\begin{aligned} R_E &= .12 + (.12 - .08) \times .25 \\ &= .13, \text{ or } 13\% \end{aligned}$$

In the second case, verify that the debt-equity ratio is 1.0, so the cost of equity is 16 percent.

We can now calculate the WACC assuming that the percentage of equity financing is 80 percent, the cost of equity is 13 percent, and the tax rate is zero:

$$\begin{aligned} \text{WACC} &= (E/V) \times R_E + (D/V) \times R_D \\ &= .80 \times .13 + .20 \times .08 \\ &= .12, \text{ or } 12\% \end{aligned}$$

In the second case, the percentage of equity financing is 50 percent and the cost of equity is 16 percent. The WACC is:

$$\begin{aligned} \text{WACC} &= (E/V) \times R_E + (D/V) \times R_D \\ &= .50 \times .16 + .50 \times .08 \\ &= .12, \text{ or } 12\% \end{aligned}$$

EXERCISE

Madhu Corporation has the following book value capital structure:

Equity capital (30 million shares, Rs.10 par)	Rs.300 million
Preference capital, 15 percent (1,000,000 shares, Rs.100 par)	Rs. 100 million
Retained earnings	Rs. 100 million
Debentures 11 percent (2,500,000 debentures, Rs.100 par)	Rs .250 million
Term loans, 13 percent	Rs. 300 million
	Rs.1050 million

The next expected dividend per share is Rs.4.00. The dividend per share is expected to grow at the rate of 15 percent. The market price per share is Rs.80. Preference stock, redeemable after 6 years, is currently selling for Rs.110 per share. Debentures, redeemable after 6 years, are selling for Rs.102 per debenture. The tax rate for the company is 35 percent.

(a) Calculate the average cost of capital using
   (i) book value proportions, and
   (ii) market value proportions

SOLUTION

(a) (i) The cost of equity and retained earnings

$$r_E = D_1/P_O + g$$
$$= 4.0 / 80 + 0.15 = 20\ \%$$

The cost of preference capital, using the approximate formula, is :

$$r_P = \frac{15 + (100-110)/6}{0.6 \times 110 + 0.4 \times 100} = 12.58\ \%$$

The pre-tax cost of debentures, using the approximate formula, is :

$$r_D = \frac{11 + (100-102)/6}{0.6 \times 102 + 0.4 \times 100} = 10.54\ \%$$

The post-tax cost of debentures is

$$10.54\ (1\text{-tax rate}) = 10.54\ (1 - 0.35)$$
$$= 6.85\ \%$$

The post-tax cost of term loans is

$$13\ (1\text{-tax rate}) = 13\ (1 - 0.35)$$
$$= 8.45\ \%$$

Source of capital	*Component Cost* (1)	*Book value Rs. in million* (2)	*Book value proportion* (3)	*Product of* (1) & (3)
Equity capital	20.00%	300	0.29	5.8
Preference capital	12.58 %	100	0.09	1.13
Retained earnings	20.00%	100	0.09	1.80
Debentures	6.85 %	250	0.24	1.64
Term loans	8.45%	300	0.29	2.45
		1050	Average cost of capital	12.82 %

Source of capital	*Component cost* (1)	*Market value Rs. in million* (2)	*Market value proportion* (3)	*Product of* (1) & (3)
Equity capital and retained earnings	20.00%	2400	0.78	15.60
Preference capital	12.58%	110	0.04	0. 50
Debentures	6.85%	255	0.08	0. 55
Term loans	8.45%	300	0.10	0. 85
		3065	Average cost of capital	17.50 %

MM THEORY WITH TAXES

- Value of firm with taxes
- EBIT to be $1,000 every year forever.
- Firm L has issued $1,000 worth of perpetual bonds on which it pays 8 percent interest each year. The interest bill is 0.08 × $1,000 = $80 every year forever.
- Also, the corporate tax rate is 21 %.

	Firm U	Firm L
EBIT	$1,000	$1,000.00
Interest	0	80.00
Taxable income	$1,000	$ 920.00
Taxes (21%)	210	193.20
Net income	$ 790	$ 726.80

INTEREST TAX SHIELD

- assume that depreciation is zero
- capital spending is zero
- there are no changes in NWC.

Cash Flow from Assets	Firm U	Firm L
EBIT	$1,000	$1,000.00
−Taxes	210	193.20
Total	$ 790	$ 806.80

Cash Flow	Firm U	Firm L
To stockholders	$790	$726.80
To bondholders	0	80.00
Total	$790	$806.80

The fact that interest is deductible for tax purposes has generated a tax savings equal to the interest payment ($80) multiplied by the corporate tax rate (21 percent): $80 x 0.21 = $16.80.

Total cash flow to firm L is $16.80 more.

Interest tax shield

The tax savings attained by a firm from interest expense.

VALUE OF TAX SHIELD

- **TAXES AND M&M PROPOSITION I**

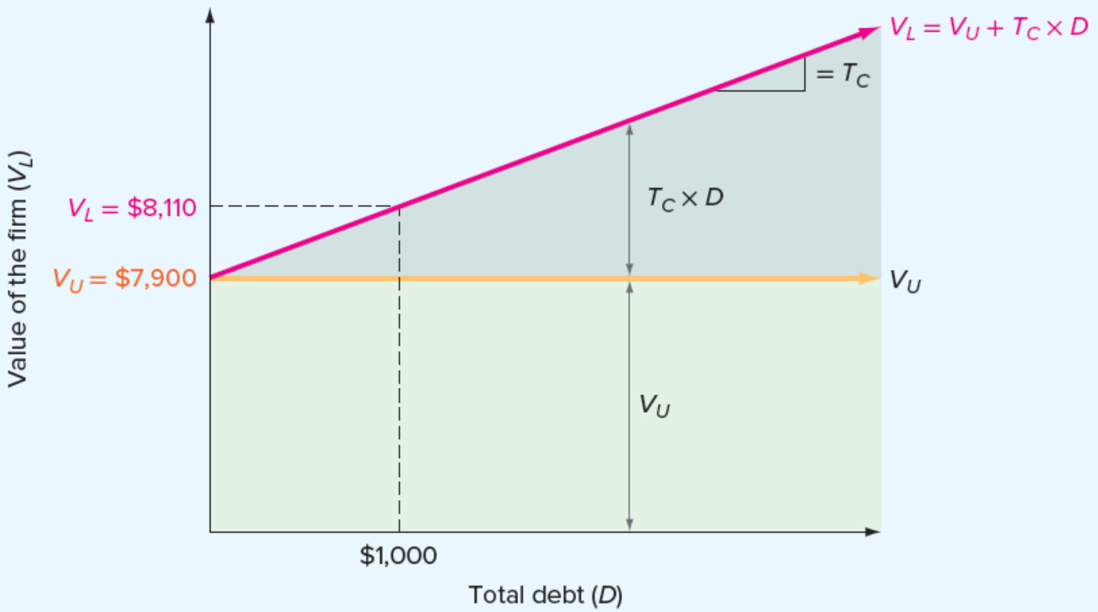
- Firm L is worth more than Firm U, the difference being the value of this $16.80 perpetuity.

$$\text{PV} = \frac{\$16.80}{.08} = \frac{.21 \times \$1{,}000 \times .08}{.08} = .21 \times \$1{,}000 = \$210$$

$$\begin{aligned}\textbf{Present value of the interest tax shield} &= (T_C \times D \times R_D)/R_D \\ &= T_C \times D\end{aligned}$$

$$V_L = V_U + T_C \times D$$

TAX EFFECT



The value of the firm increases as total debt increases because of the interest tax shield. This is the basis of M&M Proposition I with taxes.

EXAMPLE

- Suppose that the cost of capital for Firm U is 10 percent. We can call this the unlevered cost of capital, and we will use the symbol R_U to represent it. We can think of R_U as the cost of capital a firm would have if it had no debt. Firm U's cash flow is $790 every year forever, and, because U has no debt, the appropriate discount rate is $R_U = 10\%$. The value of the unlevered firm, V_U, is:

$$
\begin{aligned}
V_U &= \frac{\text{EBIT} \times (1 - T_C)}{R_U} \\
&= \frac{\$790}{.10} \\
&= \$7,900
\end{aligned}
$$

The value of the levered firm, V_L, is:

$$
\begin{aligned}
V_L &= V_U + T_C \times D \\
&= \$7,900 + .21 \times 1,000 \\
&= \$8,110
\end{aligned}
$$

<u>Firm L</u>
Debt= $1000
(perpetual bond)

Tax= 21%

MM PROPOSITION II (WITH TAXES)

- TAXES, THE WACC, AND PROPOSITION II

$$\text{WACC} = (E/V) \times R_E + (D/V) \times R_D \times (1 - T_C)$$

M&M Proposition II with corporate taxes states that the cost of equity is:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

EXAMPLE

- To illustrate, recall that we saw a moment ago that Firm L is worth \$8,110 total. Because the debt is worth \$1,000, the equity must be worth \$8,110 − 1,000 = \$7,110. For Firm L, the cost of equity is:

$$
\begin{aligned}
R_E &= .10 + (.10 - .08) \times (\$1{,}000/\$7{,}110) \times (1 - .21) \\
&= .1022, \text{ or } 10.22\%
\end{aligned}
$$

The weighted average cost of capital is:

$$
\begin{aligned}
\text{WACC} &= (\$7{,}110/\$8{,}110) \times .1022 + (\$1{,}000/\$8{,}110) \times .08 \times (1 - .21) \\
&= .0974, \text{ or } 9.74\%
\end{aligned}
$$

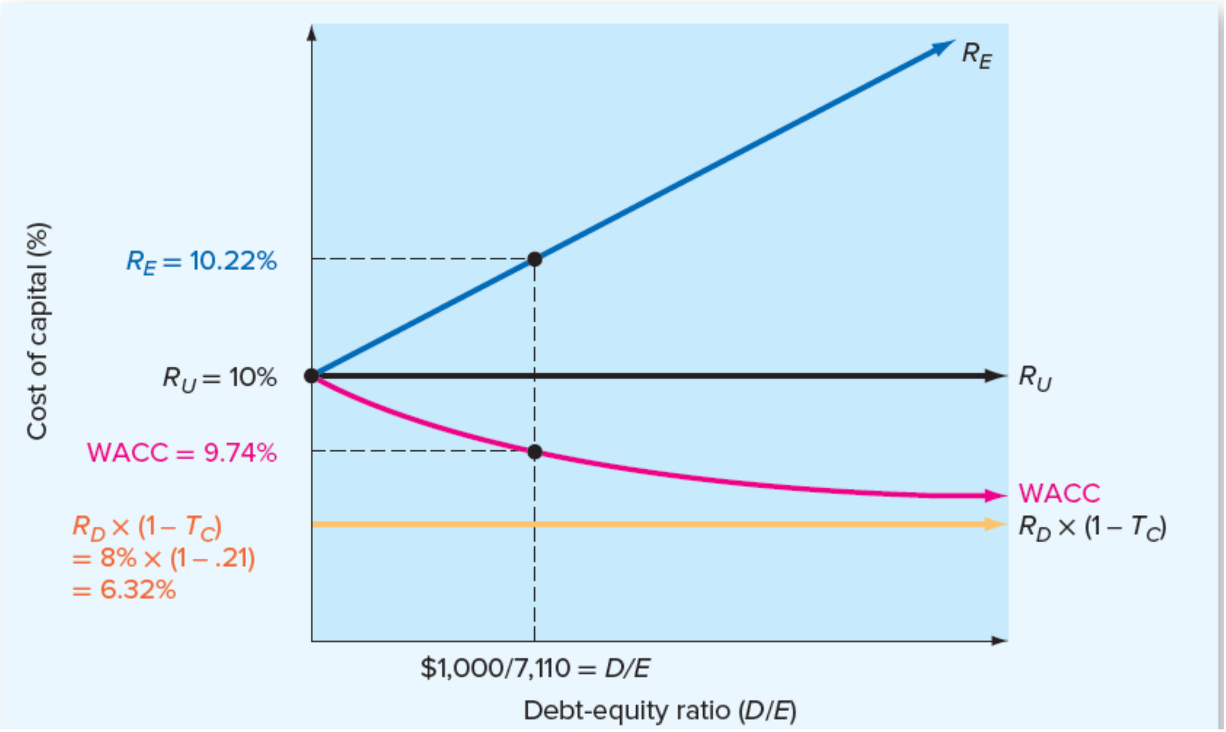
PROPOSITION II

M&M Proposition I with taxes implies that a firm's WACC decreases as the firm relies more heavily on debt financing:

$$\text{WACC} = \left(\frac{E}{V}\right) \times R_E + \left(\frac{D}{V}\right) \times R_D \times (1 - T_C)$$

M&M Proposition II with taxes implies that a firm's cost of equity, R_E, rises as the firm relies more heavily on debt financing:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$



SUMMARY MM- THEORY

I. The No-Tax Case

A. Proposition I: The value of the firm levered (V_L) is equal to the value of the firm unlevered (V_U):

$$V_L = V_U$$

Implications of Proposition I:

1. A firm's capital structure is irrelevant.
2. A firm's weighted average cost of capital (WACC) is the same no matter what mixture of debt and equity is used to finance the firm.

B. Proposition II: The cost of equity, R_E, is:

$$R_E = R_A + (R_A - R_D) \times (D/E)$$

where R_A is the WACC, R_D is the cost of debt, and D/E is the debt-equity ratio.

Implications of Proposition II:

1. The cost of equity rises as the firm increases its use of debt financing.
2. The risk of the equity depends on two things: The riskiness of the firm's operations (*business risk*) and the degree of financial leverage (*financial risk*). Business risk determines R_A; financial risk is determined by D/E.

II. The Tax Case

A. Proposition I with taxes: The value of the firm levered (V_L) is equal to the value of the firm unlevered (V_U) plus the present value of the interest tax shield:

$$V_L = V_U + T_C \times D$$

where T_C is the corporate tax rate and D is the amount of debt.

Implications of Proposition I:

1. Debt financing is highly advantageous, and, in the extreme, a firm's optimal capital structure is 100 percent debt.
2. A firm's weighted average cost of capital (WACC) decreases as the firm relies more heavily on debt financing.

B. Proposition II with taxes: The cost of equity, R_E, is:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$

where R_U is the *unlevered cost of capital*—that is, the cost of capital for the firm if it has no debt. Unlike the case with Proposition I, the general implications of Proposition II are the same whether or not there are taxes.

EXERCISE

You are given the following information for the Format Co.:

$$\text{EBIT} = \$126.58$$
$$T_C = .21$$
$$D = \$500$$
$$R_U = .20$$

The cost of debt capital is 10 percent. What is the value of Format's equity? What is the cost of equity capital for Format? What is the WACC?

SOLUTION

This one's easier than it looks. Remember that all the cash flows are perpetuities. The value of the firm if it has no debt, V_U, is:

$$V_U = \frac{\text{EBIT} - \text{Taxes}}{R_U} = \frac{\text{EBIT} \times (1 - T_C)}{R_U}$$
$$= \frac{\$100}{.20}$$
$$= \$500$$

From M&M Proposition I with taxes, we know that the value of the firm with debt is:

$$V_L = V_U + T_C \times D$$
$$= \$500 + .21 \times \$500$$
$$= \$605$$

Because the firm is worth \$605 total and the debt is worth \$500, the equity is worth \$105:

$$E = V_L - D$$
$$= \$605 - 500$$
$$= \$105$$

Based on M&M Proposition II with taxes, the cost of equity is:

$$R_E = R_U + (R_U - R_D) \times (D/E) \times (1 - T_C)$$
$$= .20 + (.20 - .10) \times (\$500/\$105) \times (1 - .21)$$
$$= .5762\text{, or } 57.62\%$$

Finally, the WACC is:

$$\text{WACC} = (\$105/\$605) \times .5762 + (\$500/\$605) \times .10 \times (1 - .21)$$
$$= .1653\text{, or } 16.53\%$$

BANKRUPTCY COST AND FIRM VALUE

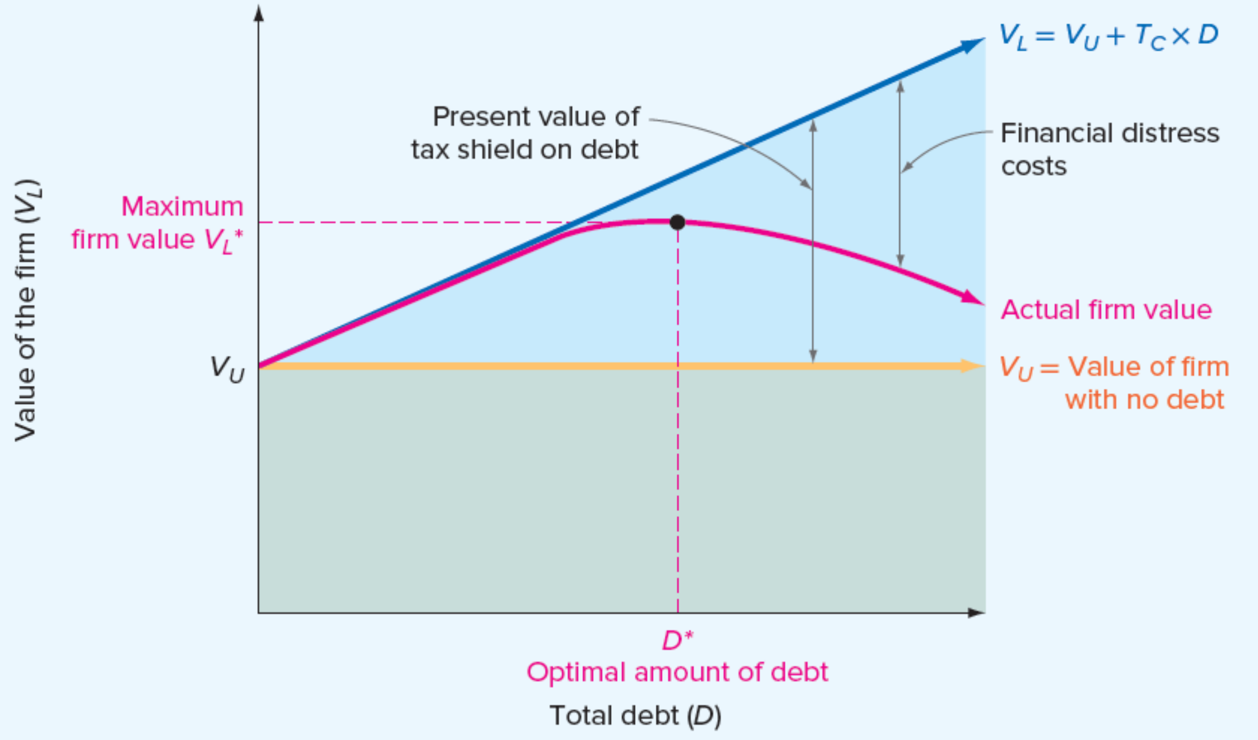
- When the value of a firm's assets equals the value of its debt, then the firm is economically bankrupt in the sense that the equity has no value.
- **DIRECT BANKRUPTCY COSTS**
  - The costs that are directly associated with bankruptcy, such as legal and administrative expenses.
  - Example: legal and administrative costs to bankruptcy
- **INDIRECT BANKRUPTCY COSTS**
  - The costs of avoiding a bankruptcy filing incurred by a financially distressed firm are called **indirect bankruptcy costs**.
- The term **financial distress costs** to refer generally to the direct and indirect costs associated with going bankrupt or avoiding a bankruptcy filing.

OPTIMAL CAPITAL STRUCTURE

- **STATIC THEORY OF CAPITAL STRUCTURE**
- It says that firms borrow up to the point where the tax benefit from an extra dollar in debt is exactly equal to the cost that comes from the increased probability of financial distress. We call this the static theory because it assumes that the firm is fixed in terms of its assets and operations and it considers only possible changes in the debt-equity ratio.

STATIC THEORY OF CAPITAL STRUCTURE

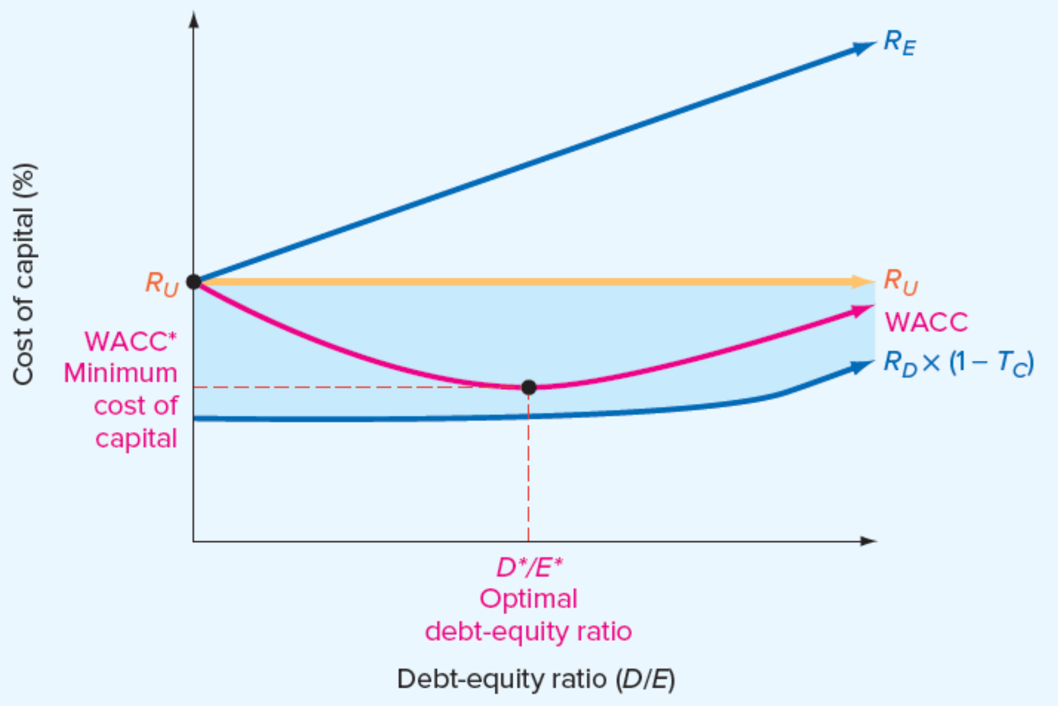
The Static Theory of Capital Structure: The Optimal Capital Structure and the Value of the Firm



According to the static theory, the gains from the tax shield on debt is offset by financial distress costs. An optimal capital structure exists that just balances the additional gain from leverage against the added financial distress cost.

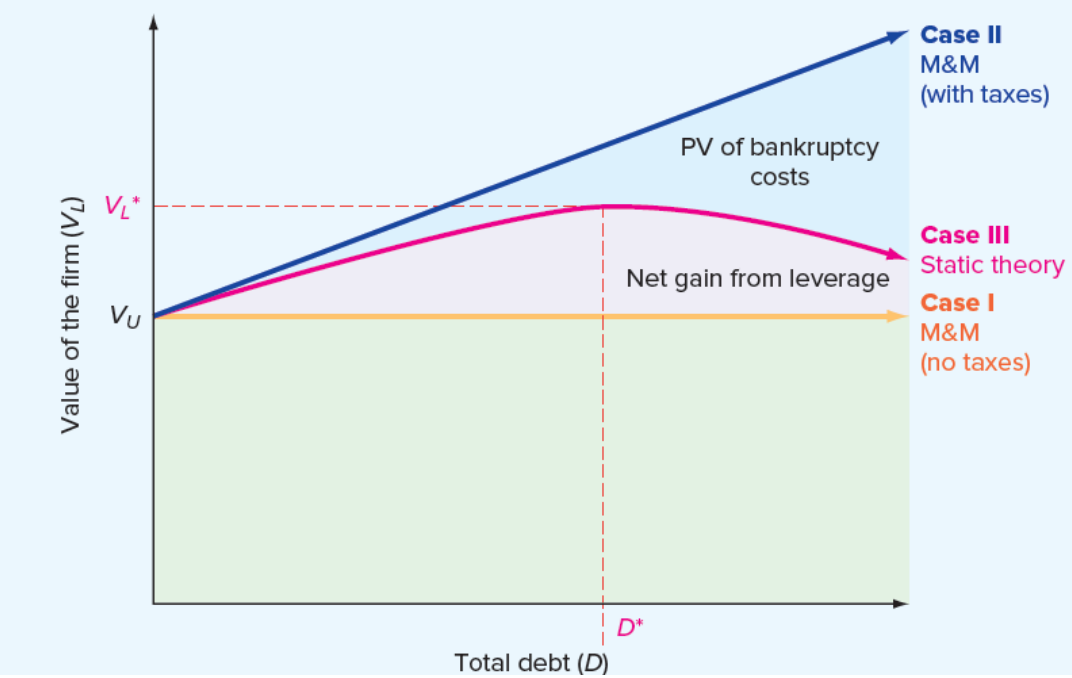
OPTIMAL CAPITAL STRUCTURE AND THE COST OF CAPITAL

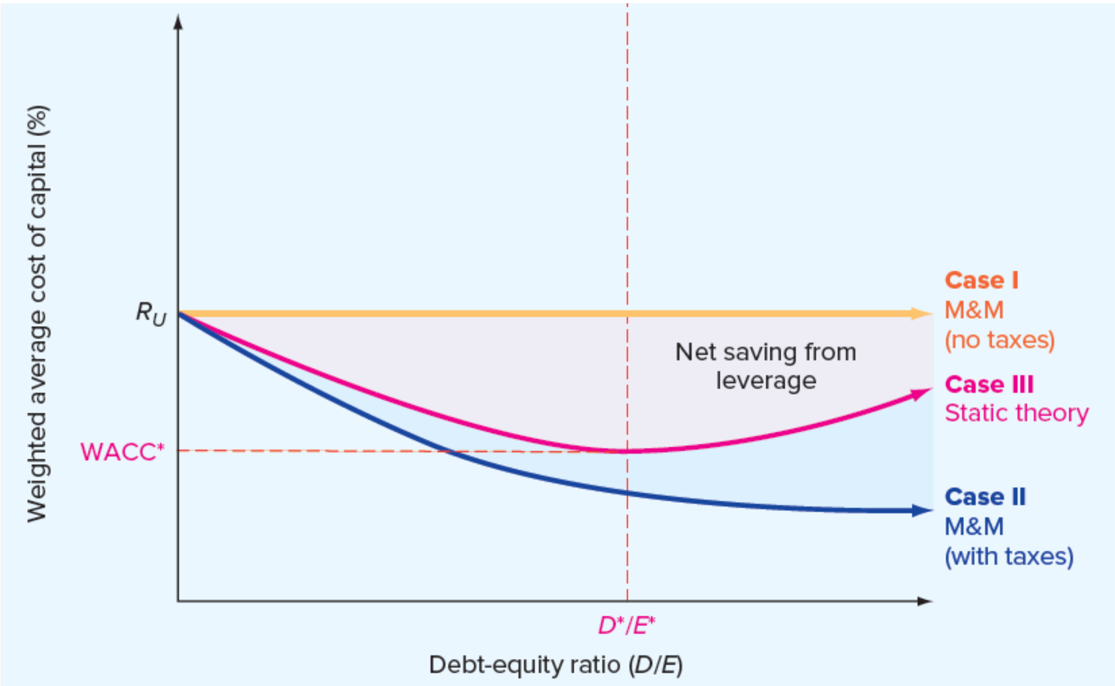
The Static Theory of Capital Structure: The Optimal Capital Structure and the Cost of Capital



According to the static theory, the WACC falls initially because of the tax advantage of debt. Beyond the point D^*/E^*, it begins to rise because of financial distress costs.

SUMMARY- MM THEORY





PECKING ORDER THEORY

- The pecking-order theory is an alternative to the static theory. A key element in the pecking-order theory is that firms prefer to use internal financing whenever possible.
- The pecking order theory suggests- Companies will use
  - internal financing first. Then, they will
  - issue debt if necessary.
  - Equity will be sold pretty much as a last resort.

IMPLICATIONS OF THE PECKING ORDER

- ***No target capital structure:*** Under the pecking-order theory, there is no target or optimal debt-equity ratio.
- ***Profitable firms use less debt:*** Because profitable firms have greater internal cash flow, they will need less external financing and will therefore have less debt. As we mentioned earlier, this is a pattern that we seem to observe, at least for some companies.
- ***Companies will want financial slack:*** To avoid selling new equity, companies will want to stockpile internally generated cash. Such a cash reserve is known as financial slack.

APPLICATION OF CAPITAL STRUCTURE

- How are operating leverage and financial leverage similar?
- How are they different?

OPERATING AND FINANCIAL LEVERAGE

- Operating Leverage
  - Involves the use of assets having fixed costs
- Financial Leverage
  - Involves the use of liabilities (and preferred stock) having fixed costs

DEGREE OF OPERATING LEVERAGE

- Degree of Operating Leverage
  - Measured as a percentage change in EBIT resulting from a given percentage change in sales

$$\text{DOL at X} = \frac{\dfrac{\Delta \text{EBIT}}{\text{EBIT}}}{\dfrac{\Delta \text{Sales}}{\text{Sales}}} \text{ or } \frac{\text{Sales - Variable Costs}}{\text{EBIT}}$$

EXAMPLE

Effect on Earnings per Share of a 10 Percent Increase in Sales, Allegan Manufacturing Company, Year Ending December 31, 2016

		(1)		(2)	Percentage Change [(2) − (1)] ÷ (1)
Sales		$ 5,000,000		$ 5,500,000	+10%
Less: Variable operating costs (0.60 × Sales)	$3,000,000		$3,300,000		+10
Fixed operating costs	1,000,000		1,000,000		0
Total operating costs		4,000,000		4,300,000	+8
Earnings before interest and taxes		$ 1,000,000		$ 1,200,000	+20
Less: Interest payments (fixed capital cost)		250,000		250,000	0
Earnings before taxes		$ 750,000		$ 950,000	+27
Less: *Income taxes* (variable), 40%		300,000		380,000	+27
Earnings after taxes		$ 450,000		$ 570,000	+27
Less: Preferred dividends (fixed capital cost)		150,000		150,000	0
Earnings available to common stockholders		$ 300,000		$ 420,000	+40
Earnings per share (100,000 shares)		$ 3.00		$ 4.20	+40

DOL CALCULATION

$$\text{DOL} = \frac{Q(P-V)}{Q(P-V)-F}$$

Q = the number of units;
P = the price per unit;
V = the variable operating cost per unit;
F = the fixed operating cost;
P – V = the per unit contribution margin; and
Q (P – V) = the contribution margin

$$\text{DOL at X} = \frac{\frac{\Delta\text{EBIT}}{\text{EBIT}}}{\frac{\Delta\text{Sales}}{\text{Sales}}} \text{ or } \frac{\text{Sales - Variable Costs}}{\text{EBIT}}$$

$$\begin{aligned}\text{DOL at } \$5{,}000{,}000 &= \frac{\frac{(\$1{,}200{,}000 - \$1{,}000{,}000)}{\$1{,}000{,}000}}{\frac{(\$5{,}500{,}000 - \$5{,}000{,}000)}{\$5{,}000{,}000}} \\ &= \frac{\$200{,}000}{\$1{,}000{,}000} \times \frac{\$5{,}000{,}000}{\$500{,}000} \\ &= 2.0\end{aligned}$$

DOL AT VARIOUS SALES LEVEL

DOL at Various Sales Levels, Allegan Manufacturing Company

Sales, TR = P × Q*	Degree of Operating Leverage (DOL)
$ 500,000	−0.25
1,000,000	−0.67
1,500,000	−1.50
2,000,000	−4.00
2,500,000**	(Undefined)
3,000,000	+6.00
3,500,000	+3.50
4,000,000	+2.67
4,500,000	+2.25
5,000,000	+2.00
5,500,000	+1.83
6,000,000	+1.71

*TR = Total revenue, P = Price, Q = Quantity

DEGREE OF FINANCIAL LEVERAGE

- Measurement of Operating and Financial Leverage
  - Degree of Financial Leverage
    - Measured as the percent change in EPS resulting from a given percent change in EBIT

$$\text{DFL at X} = \frac{\dfrac{\Delta \text{EPS}}{\text{EPS}}}{\dfrac{\Delta \text{EBIT}}{\text{EBIT}}} \quad \text{or} \quad \frac{\text{EBIT}}{\text{EBIT} - \text{I} - \dfrac{D_p}{(1-T)}}$$

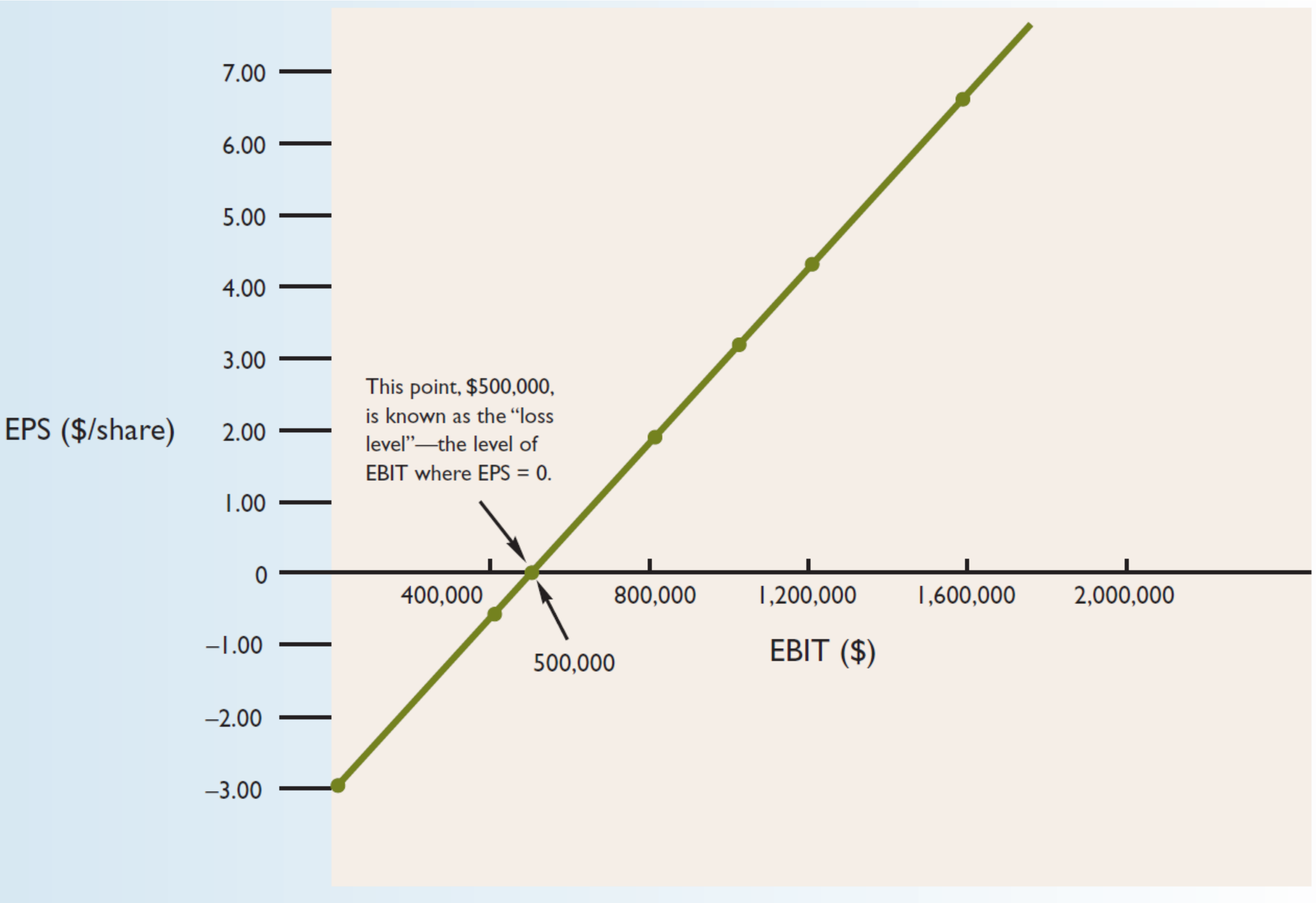
D_p is firm’s preferred dividend payments

$$\text{DFL at } \$1{,}000{,}000 = \frac{\dfrac{(\$4.20 - \$3.00)}{\$3.00}}{\dfrac{(\$1{,}200{,}000 - \$1{,}000{,}000)}{\$1{,}000{,}000}}$$

$$= \frac{\$1.20}{\$3.00} \times \frac{\$1{,}000{,}000}{\$200{,}000}$$

$$= 2.0$$

EPS-EBIT



7.00
6.00
5.00
4.00
3.00
2.00
1.00
0
–1.00
–2.00
–3.00
EPS ($/share)
This point, $500,000, is known as the "loss level"—the level of EBIT where EPS = 0.
400,000
800,000
1,200,000
1,600,000
2,000,000
500,000
EBIT ($)

DEGREE OF COMBINED LEVERAGE

- Degree of Combined Leverage
  - Measured as a percentage change in EPS resulting from a given percentage change in sales

$$\text{DCL at X} = \frac{\dfrac{\Delta \text{EPS}}{\text{EPS}}}{\dfrac{\Delta \text{Sales}}{\text{Sales}}} \text{ or } \frac{\text{Sales - Variable Costs}}{\text{EBIT - I - }\dfrac{D_p}{(1\text{-T})}} \text{ or DOL} \times \text{DFL}$$

DCL

$$\text{DCL at } \$5{,}000{,}000 = \frac{\frac{(\$4.20 - \$3.00)}{\$3.00}}{\frac{(\$5{,}500{,}000 - \$5{,}000{,}000)}{\$5{,}000{,}000}}$$

$$= \frac{\$1.20}{\$3.00} \times \frac{\$5{,}000{,}000}{\$500{,}000}$$

$$= 4.0$$

$$\text{DCL at } \$5{,}000{,}000 = \frac{\$5{,}000{,}000 - \$3{,}000{,}000}{\$1{,}000{,}000 - \$250{,}000 - \$150{,}000/(1 - 0.40)}$$

$$= 4.0$$